

THE AI ARTIFACT LEXICON

A Complete Forensic Reference Guide to AI-Generated Image Flaws, Hallmarks,
and Detection Methodology

Version 1.1 · April 2025 — Corrected Edition

For use with: AuthenticLens — The Realism Architect Gemini Gem
Confidential Reference Document

HOW TO USE THIS LEXICON

This lexicon is organised into nine major artifact categories, each subdivided into specific flaw types. For each artifact, you will find: (1) a precise technical definition, (2) the underlying generation mechanism, (3) visual detection markers, (4) spatial regions where the artifact most commonly appears, and (5) confidence-weighting implications for the Phase 1 AI detection score.

When analysing an image, cross-reference findings from multiple categories. A single artifact carries low individual weight; the convergence of three or more artifact types across different categories raises the AI confidence score significantly. Use the Composite Scoring Table at the end of this document to calculate the final verdict.

■ ERRATA — GLOBAL CORRECTION

The original scoring system treated smoothness, glow, and perfect skin as strong primary AI evidence. This is corrected throughout this edition. Editorial tools (Adobe Lightroom, Camera Raw, Photoshop) and computational phone-camera pipelines can produce identical surface effects on real photographs. Structural and geometric failures are always weighted above stylistic cues.

CATEGORY 1 — ANATOMICAL ANOMALIES

Anatomical anomalies represent the most well-documented class of AI image artifacts. They arise because AI image generators do not understand human anatomy as a coherent structural system. Instead, models learn statistical correlations between pixels, approximating the appearance of body parts rather than

generating them from a structural blueprint.

1.1 — HAND AND FINGER DEFORMITIES

■ ERRATA — Doc 1 § 1.1	ORIGINAL TEXT stated: 'Hands are the single most reliable indicator of AI generation.' This is DOWNGRADED. Modern generative architectures (Midjourney v6, Imagen 3, Gemini image models) have substantially resolved gross anatomical fusions. Hands are a high-weight cue but must be used alongside text fidelity, lighting logic, material physics, and scene logic. Motion blur, aggressive noise reduction, and low-light smearing can distort authentic hands.
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Hands remain a high-value forensic zone. The human hand contains 27 bones, 29 joints, and over 30 muscles. AI models systematically approximate rather than construct this structure. Primary indicators:

- **DIGIT COUNT ERRORS:** Six, seven, or four fingers per hand. Count every visible digit methodically.
 - **FINGER LENGTH ANOMALIES:** Incorrect taper, disproportionate length, or abrupt termination without proper fingertip or nail structure.
 - **JOINT FUSION:** Knuckles appearing fused or missing, producing fingers that bend as uniform tubes.
 - **PALM TOPOLOGY ERRORS:** Implausibly wide palms, extra thenar (thumb pad) or hypothenar (pinky pad) eminences, or incorrect metacarpal ridges.
 - **THUMB POSITIONING:** Thumb at anatomically impossible angle, or two thumbs on one hand.
 - **NAIL MALFORMATION:** Nails absent, on the wrong surface, extending implausibly, or duplicated.
 - **WRIST DISCONTINUITY:** Structural break where the wrist does not logically connect to the palm.
 - **SKIN WEBBING:** Excess webbing between fingers, particularly index-middle and ring-pinky pairs.
- *NOTE: Examine all hands at 200% zoom minimum. Hands near image edges, partially occluded hands, and hands interacting with objects are highest-risk zones. A hand with five correctly-proportioned fingers still requires scrutiny — examine the palmar surface texture and thumb web-space.*

1.2 — FACIAL SYMMETRY AND MICROSTRUCTURE ERRORS

- **PUPIL ASYMMETRY (PATHOLOGICAL TYPE):** Unlike natural minor asymmetry, AI eyes frequently show one pupil significantly larger or differently shaped, or pupils not centred within the iris.
- **IRIS DETAIL COLLAPSE:** Real irises contain complex trabecular meshwork patterns. AI irises often show generic radiating lines or a uniform gradient.
- **CATCHLIGHT DUPLICATION:** Multiple catchlights or catchlights in geometrically impossible positions relative to stated lighting.
- **SCLERA SMOOTHNESS:** Unnaturally uniform white sclera lacking subtle redness and episcleral vessels.
- **HELIX AND ANTIHELIX ERRORS:** Complex cartilaginous ear ridges frequently simplified, merged, or given incorrect topological relationships.
- **EAR LOBE ATTACHMENT:** Free vs attached earlobes generated without structural logic; transition from lobe to jawline is a reliable detection point.
- **BILATERAL SYMMETRY OVERCORRECTION:** Real ears on the same face differ slightly; AI ears are often mirror images.

- **TOOTH COUNT AND SPACING:** AI teeth may exceed anatomically correct count, or show unnaturally perfect spacing and uniform sizing.
- **GINGIVAL DETAIL FAILURE:** Gum line, gingival papillae, and natural colour variation rarely reproduced correctly.
- **TONGUE SURFACE TEXTURE:** Real tongue papillae create specific texture absent in AI, which tends to produce a smooth, uniformly pink surface.

1.3 — HAIR GENERATION ARTIFACTS

- **HAIRLINE DISSOLUTION:** Transition from scalp to hair is a common failure point — AI hair frequently melts into the scalp without individual strand origins.
 - **STRAND COHERENCE FAILURE:** AI hair may contain strands that reverse direction without mechanical cause.
 - **HAIR-BACKGROUND MERGING:** Individual hairs at the silhouette edge blend into the background, manifesting as a halo artifact.
 - **TEXTURE UNIFORMITY:** AI hair often shows uniformly distributed specular highlights, creating an implausibly consistent sheen.
 - **FACIAL HAIR TOPOLOGY:** Beards and moustaches frequently disconnect from the face at jaw, under-nose, or sideburn-to-beard junctions.
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CATEGORY 2 — TEXT AND TYPOGRAPHY ARTIFACTS

Text rendering is one of the most definitive AI failure categories. Diffusion models do not have a discrete 'render text' function — they generate text as a visual pattern statistically associated with letter-like shapes. The result is pseudo-text that mimics the appearance of letters without encoding semantic content.

2.1 — PSEUDO-ALPHABETIC GENERATION

- **LETTER MORPHING:** Individual letters blend into adjacent letters, sharing strokes or merging counters.
- **IMPOSSIBLE LIGATURES:** Connections between letters that do not exist in any known typeface.
- **MIRRORED AND ROTATED GLYPHS:** Characters mirrored or rotated within an otherwise horizontal text line.
- **NUMERICAL CONFLATION:** Digits 3, 8, and 6 frequently interchanged or blended. Prices, dates, and addresses almost always fail on close examination.
- **SERIF/SANS-SERIF MIXING:** Serifs applied inconsistently within a single word.
- **INCONSISTENT BASELINE:** Letters drift above and below the baseline.
- **CHARACTER REPETITION ARTIFACTS:** Same correctly formed character appearing multiple times within a word.

■ **NOTE:** Zoom to 300% — at least one glyph-level error is almost always present. Examine every enclosed space (counter) in letters like a, b, d, e, g, o, p, q for proper closure.

2.2 — SIGN AND LABEL ARTIFACTS

- **STORE SIGN GARBLING:** Business names on storefronts are nearly universally unreadable in AI street photography.
 - **LABEL PSEUDO-TEXT:** Product packaging features text-like patterns without actual words.
 - **NEWSPAPER/BOOK BODY TEXT:** Body text collapses into squiggles — visually simulating text columns without rendering characters.
 - **DIGITAL DISPLAY ERRORS:** Clocks, scoreboards, and phone screens show impossible time displays or inconsistent digit counts.
 - **LOGO RECONSTRUCTION FAILURE:** Known brand logos approximately reproduced with modified proportions, colour shifts, or corrupted letterforms.
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CATEGORY 3 — INTRICATE PATTERN AND TEXTURE DISTORTIONS

Complex repeating patterns represent a fundamental limitation of diffusion-based models. Patterns requiring strict geometric repetition are generated statistically, producing local approximations that fail globally.

3.1 — CHAINMAIL AND INTERLOCKING STRUCTURES

- **RING COUNT INCONSISTENCY:** Chainmail rings vary in size without cause.
- **INTERLOCKING FAILURE:** Rings appear to overlap where they should interlock, creating a flat-print appearance.
- **PATTERN DRIFT:** The repeating unit of the weave gradually shifts direction across the image.
- **SHADOW INCONSISTENCY:** Shadows cast by individual links do not correspond to a coherent light source.

3.2 — WOVEN FABRICS AND TEXTILES

- **WARP/WEFT FAILURE:** Over-under weaving pattern breaks down — threads appear to pass over multiple threads where they should pass under, or simply merge.
- **PLAID AND CHECK PATTERN DRIFT:** Plaid colour bands converge, diverge, or change width without structural basis.
- **KNIT STRUCTURE COLLAPSE:** Knitted fabric's V-shaped stitch pattern degenerates at medium distances and dissolves at close range.
- **TEXTURE REPETITION ARTIFACTS:** Tiled textures show visible repetition seams.

3.3 — ARCHITECTURAL AND STRUCTURAL PATTERNS

- **BRICK PATTERN ERRORS:** Bricks of inconsistent size, mortar joints varying in width irrationally, course lines converging.
- **FENCE AND RAILING DISTORTIONS:** Fence pickets merge with each other, have inconsistent spacing, or curve without structural cause.
- **WINDOW GRID FAILURES:** Pane counts differing between top and bottom halves, frames not closing completely.

- FLOOR TILE PERSPECTIVE COLLAPSE: Receding tile grids losing their vanishing point or having inconsistently sized tiles.

3.4 — NATURAL REPEATING STRUCTURES

- LEAF VEIN NETWORKS: Veins that terminate without branching, branch at anatomically impossible angles, or create closed loops not connecting to the midrib.
 - CROWD SCENES: Faces and body parts beginning to merge in crowd backgrounds — arms of one figure continuing as the torso of another.
 - ANIMAL FUR PATTERNS: Bilateral symmetry of striped or spotted animals violated; stripes or spots terminating at the dorsal midline.
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CATEGORY 4 — BACKGROUND MERGING AND OBJECT BOUNDARY FAILURES

4.1 — SUBJECT-BACKGROUND BLENDING

- EDGE CONTAMINATION: The outline of the primary subject takes on the colour or texture of the adjacent background.
- TRANSPARENT SUBJECT ZONES: Geometrically complex regions allow background to bleed through in ways that do not correspond to actual transparency.
- BACKGROUND EXTENSION THROUGH OBJECTS: Background patterns continue faintly through solid opaque foreground objects.
- COLOUR HALO ARTIFACT: A subtle coloured fringe appears around subject edges, expressing the model's uncertainty about object boundary location.

4.2 — OBJECT PHYSICS AND SUPPORT FAILURES

- FLOATING OBJECTS: Items resting on surfaces without contact shadows, or appearing to float slightly above their supporting surface.
- SHADOW DIRECTION INCONSISTENCY: Multiple objects casting shadows in different directions, indicating multiple inconsistent light source assumptions.
- REFLECTION MISMATCH: Reflective surfaces reflecting objects not present in the scene, or failing to reflect objects directly in line with reflection geometry.
- GLASS AND LIQUID PHYSICS: Glass objects with incorrect refraction; liquid surfaces not showing expected Snell's Law distortion.
- CAST SHADOW SHAPE ERRORS: Shadows not accurately reproducing the silhouette of the casting object.

■ **NOTE:** *Identify all light sources in the image, then verify every shadow direction and every reflective surface is geometrically consistent with those sources.*

CATEGORY 5 — LIGHTING ARTIFACTS AND THE AI GLOW PHENOMENON

■ ERRATA — Doc 1 § 5 CORRECTION

ORIGINAL SCORING treated smooth skin + AI glow as primary AI evidence worth 15 points. This is REVISED. Editorial tools (Lightroom clarity, texture, skin-smoothing), phone-camera computational pipelines, beauty retouching, and denoise processing all create smooth skin and strong specular highlights on real photographs. These surface effects must NOT be primary AI evidence. They may contribute only when structural failures are also present.

5.1 — HYPER-SMOOTH DIFFUSE LIGHTING

- PORE ELIMINATION: Real skin photographed under natural light shows pores, fine lines, and subsurface scattering variations. NOTE: editorial smoothing and phone HDR denoise can create this effect on real photos.
- ABSENCE OF SKIN MICRO-RELIEF: Faint elevation variations of real skin (follicle patterns, vellus hair impressions, dermal ridges) are absent in AI.
- UNIFORM SUBSURFACE SCATTERING: AI skin applies a uniform SSS approximation rather than the real variation where earlobes, nasal tips, and finger web-spaces appear warmer and more translucent.
- SPECULAR HIGHLIGHT PERFECTION: Highlights on skin, hair, and objects are geometrically perfect in a way that real specular reflections, subject to surface micro-roughness, never are.

5.2 — LIGHT SOURCE IMPOSSIBILITY

- OMNIDIRECTIONAL KEY LIGHT: Subject lit from a direction requiring the light source to be inside or behind the camera, yet the background shows no evidence of this light.
- MULTIPLE INCONSISTENT SOURCES: Face lit from camera-left on the left cheek and simultaneously from camera-right on the right cheek.
- LIGHT WITHOUT ORIGIN: Environmental lights visible in the frame not producing fall-off or beam patterns consistent with their stated intensity and position.
- INVERSE-SQUARE LAW VIOLATIONS: Implausibly uniform brightness across distances where real light would have fallen off significantly.

5.3 — HYPER-SATURATION AND COLOUR PERFECTION

- EXCESSIVE SATURATION: Colours perceptually saturated beyond what film or digital sensors would capture under normal conditions.
 - HUE UNIFORMITY: Real photographs contain complex hue variation within nominally single-colour objects. AI objects often render with artificially uniform hues.
 - ATMOSPHERIC DESATURATION FAILURE: AI backgrounds frequently maintain full saturation at distances where real images would show aerial perspective (desaturation and blue shift).
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CATEGORY 6 — MATERIAL PHYSICS AND SURFACE PROPERTY FAILURES

6.1 — METALLIC SURFACE ERRORS

- **ANISOTROPIC REFLECTION FAILURE:** Brushed metals reflect light in elongated highlights aligned with the brushing direction. AI metals produce isotropic (circular) highlights regardless of surface treatment.
- **CHROME AND MIRROR SURFACE DISTORTION:** Highly reflective surfaces should show a distorted wide-angle reflection of the surrounding environment. AI chrome frequently shows a generic gradient or non-contextual reflection.
- **OXIDATION AND PATINA ABSENCE:** Real metals in environmental conditions show patina and oxidation. AI metals are implausibly pristine.

6.2 — GLASS AND TRANSPARENT MATERIAL ERRORS

- **INTERNAL REFLECTION ABSENCE:** Real glass shows faint internal reflections (ghost images) caused by light bouncing between parallel surfaces.
- **EDGE CAUSTIC FAILURE:** Glass edges concentrate light into caustic patterns on adjacent surfaces. AI glass does not generate caustics.
- **THICKNESS-DEPENDENT TINT:** Thick glass develops a characteristic green or blue tint at edges. AI glass maintains uniform clarity regardless of apparent thickness.

6.3 — FABRIC AND CLOTHING PHYSICS

- **GRAVITY-DEFYING DRAPE:** Fabric drapes and folds according to weight, stiffness, and attachment points. AI fabric frequently shows drape patterns inconsistent with apparent fabric weight or gravity direction.
 - **CREASE AND WRINKLE LOGIC:** Real clothing shows creases originating from articulation points (elbow, knee, crotch). AI wrinkles often appear at random locations without biomechanical cause.
 - **POCKET BULGE ABSENCE:** Clothing containing objects shows characteristic deformation at the pocket location. AI clothing pockets rarely show this bulge.
 - **SEAM AND STITCH DETAIL:** Real clothing shows stitching lines, seam allowances, and construction details typically simplified or absent in AI garments.
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CATEGORY 7 — STRUCTURAL AND SPATIAL COHERENCE FAILURES

7.1 — PERSPECTIVE AND VANISHING POINT ERRORS

- **MULTI-PERSPECTIVE FUSION:** AI images of architectural interiors or street scenes sometimes fuse two different camera perspectives, creating impossible spatial geometry where receding lines converge to multiple inconsistent vanishing points.
- **SCALE INCONSISTENCY:** Objects in the same apparent depth plane vary in size inconsistently with a unified perspective projection.

- HORIZON LINE DISCONTINUITY: The implied horizon line shifts between foreground and background zones of the same image.

7.2 — SPATIAL CONTINUITY FAILURES

- BODY PART CONTINUITY BREAKS: Limbs occasionally severed or disconnected at mid-point, reappearing at the wrong position.
- OBJECT DUPLICATION: Two shadows for the same object, two watch faces on a wristwatch, or a reflected element appearing also in the unreflected foreground.
- ROOM TOPOLOGY ERRORS: Indoor environments containing structural impossibilities — a wall continuing on both sides of a window without the window creating a proper opening.

CATEGORY 8 — MODEL-SPECIFIC VISUAL SIGNATURES

Each major AI image generation platform has characteristic visual signatures — not flaws per se, but aesthetic tendencies that become recognisable with exposure. These signatures are useful secondary indicators only when combined with artifact detection. Do not lead with model signatures.

Model	Characteristic Signature	Common Artifacts	Confidence Use
Midjourney v6	Hyper-detailed, painterly realism; dramatic rim lighting; oversaturated skies	Skin over-smoothing; fabric weave errors; background merging at image edges	Secondary — combine with structural evidence
DALL-E 3	Cleaner, more illustrative aesthetic; better text than v2; flatter lighting	Hand digit errors persist; text garbled in complex layouts; logo reconstruction failures	Moderate — harder to identify in isolation
Stable Diffusion (Base)	Variable by checkpoint; base models show visible noise at low step counts	Anatomical errors more frequent; face topology failures; background hallucinations	Higher — noise artifacts visible
Adobe Firefly	Stock-photo aesthetic; overly clean composition; corporate-safe colour palette	Extremely smooth skin; generic expressions; background artifacting at subject edges	High if subject-background edge is too clean
Imagen / Gemini Image Models	High photorealism target; often indistinguishable at low zoom	Microstructure failures at 200%+ zoom; subtle hand errors; reflection inconsistency	Low — requires close examination

CATEGORY 9 — PHASE 1 COMPOSITE CONFIDENCE SCORING SYSTEM

■ ERRATA — SCORING REVISION	The original scoring system has been revised. (1) Smooth skin + AI glow score has been REMOVED as a standalone primary indicator. (2) An EDITORIAL OVERRIDE of -20 points applies when flagged features are better explained by computational photography, Lightroom-style editing, or beauty retouching, AND no structural or geometric failures are present. (3) The system now integrates with the four-way classification of Document 4.
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Artifact Category	Weight	Scoring Notes
Hand/Finger Deformity (structural only)	20 pts	High-weight but conditional. Verify motion blur, denoise, and low-light smearing are not the cause. Gross anatomical fusions only.
Text/Typography Failure	20 pts	Any pseudo-alphabetic generation in visible text elements. Strongest structural evidence.
Eye Anatomy Error	15 pts	Pupil asymmetry (pathological type), iris flatness, or catchlight duplication.
Background Merging / Edge Bleeding	10 pts	Visible at subject-background boundaries in ways not explained by portrait-mode masking.
Object Physics Violation	10 pts	Floating objects, shadow direction error, or reflection mismatch.
Pattern/Texture Distortion	10 pts	Structural breakdown in repeating patterns (weave, brick, railing).
Structural/Perspective Error	8 pts	Vanishing point inconsistency or spatial impossibility.
Material Property Failure	7 pts	Metal, glass, or fabric physics errors.
AI Glow / Smooth Skin (surface only)	0–5 pts	Reduced from 15 pts. Only score if structural failures are also present. Zero points if editorial processing explains the look.
Model Aesthetic Signature	5 pts	Secondary indicator only. Do not lead with this.
EDITORIAL OVERRIDE	–20 pts	Apply if flagged features are better explained by Lightroom, phone HDR, or beauty retouching, AND no structural failures exist.

SCORE INTERPRETATION

Score Range	Verdict	Recommended Action
0–20	HIGH PROBABILITY AUTHENTIC	Proceed with standard realism enhancement only
21–40	INCONCLUSIVE	Flag for secondary review; apply selective realism corrections
41–65	PROBABLE AI	Apply full Phase 2 modification protocol

Score Range	Verdict	Recommended Action
66–85	HIGH CONFIDENCE AI	Full Phase 2 required; note specific primary artifacts
86–100	DEFINITIVE AI	Multiple high-weight artifacts confirmed; immediate Phase 2